

Behavior-Based HTTP Anomaly Detection and Incident Response using Wazuh, OpenSearch ML, Slack, and TheHive

This project demonstrates a **real-world SOC (Security Operations Center) workflow** where **machine learning-based anomaly detection** is used to identify abnormal HTTP error behavior on a web server. The detected anomaly is **alerted in real time via Slack**, investigated using **OpenSearch Discover**, mapped to **MITRE ATT&CK**, escalated into **TheHive as an incident case**, and finally **mitigated by blocking the attacker at the host level**.

PROJECT OBJECTIVES

- Detect **abnormal HTTP error patterns** using ML-based anomaly detection
- Generate **real-time alerts** for anomalous behavior
- Investigate logs and attacker activity
- Map findings to **MITRE ATT&CK**
- Perform **incident response** using TheHive
- Apply **host-based mitigation** and verify effectiveness

ENVIRONMENT & TOOLS USED

- **Wazuh** – Log collection & rule-based detection
- **OpenSearch Anomaly Detection (ML)** – Behavior-based detection
- **Apache Web Server** – Target application
- **Kali Linux** – Attacker simulation
- **Slack** – Real-time alert notifications
- **TheHive** – Incident response & case management
- **MITRE ATT&CK Framework** – Threat mapping

WHY ANOMALY DETECTION?

Traditional rule-based detection triggers only on **known patterns**.

Anomaly detection identifies **unusual behavior**, such as:

- Sudden spikes in HTTP 404 / 400 errors
- Abnormal request rates
- Reconnaissance or scanning activity

Benefits:

- Detects **unknown or zero-day attack patterns**
- Reduces dependence on static rules
- Learns **baseline behavior** automatically

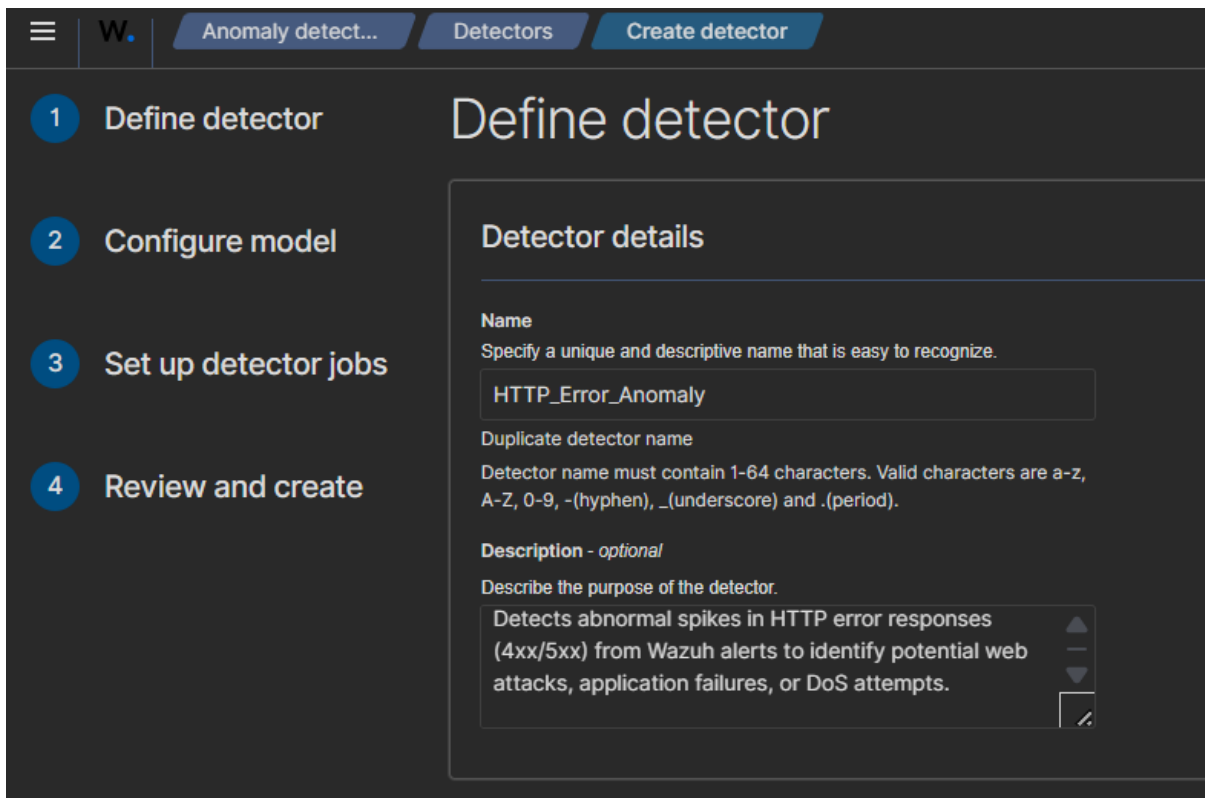
STEP 1: ANOMALY DETECTOR CONFIGURATION

What is being done:

- Created an **Anomaly Detector** in OpenSearch
- Selected index: wazuh-alerts-*
- Applied filter: rule.groups = web
- Timestamp field: @timestamp

Feature configured:

- **Feature name:** http_error_count
- **Aggregation:** value_count
- **Field:** rule.id
- **Interval:** 1 minute
- **Shingle size:** 8



The screenshot shows the 'Define detector' configuration page in OpenSearch. The page has a dark theme and a sidebar on the left with four steps: 1. Define detector (active), 2. Configure model, 3. Set up detector jobs, and 4. Review and create. The main content area is titled 'Define detector' and contains a 'Detector details' section. This section has two fields: 'Name' and 'Description - optional'. The 'Name' field is labeled 'Name' and has a description 'Specify a unique and descriptive name that is easy to recognize.' The value entered is 'HTTP_Error_Anomaly'. Below the name field, there is a warning: 'Duplicate detector name' and a note: 'Detector name must contain 1-64 characters. Valid characters are a-z, A-Z, 0-9, -(hyphen), _(underscore) and .(period).' The 'Description - optional' field is labeled 'Description - optional' and has a description 'Describe the purpose of the detector.' The value entered is 'Detects abnormal spikes in HTTP error responses (4xx/5xx) from Wazuh alerts to identify potential web attacks, application failures, or DoS attempts.' There are up and down arrow icons to the right of the description text, and a small icon in the bottom right corner of the description box.

W. Anomaly detect... Detectors Create detector

1 Define detector

2 Configure model

3 Set up detector jobs

4 Review and create

Define detector

Detector details

Name
Specify a unique and descriptive name that is easy to recognize.

HTTP_Error_Anomaly

Duplicate detector name
Detector name must contain 1-64 characters. Valid characters are a-z, A-Z, 0-9, -(hyphen), _(underscore) and .(period).

Description - optional
Describe the purpose of the detector.

Detects abnormal spikes in HTTP error responses (4xx/5xx) from Wazuh alerts to identify potential web attacks, application failures, or DoS attempts.

Select Data

Clusters

Select a local cluster or remote clusters from cross-cluster connections.

wazuh-cluster (Local) ×

Index

Choose an index, index pattern or alias as the data source.

wazuh-alerts-* ×

You can use a wildcard (*) in your index pattern.

Data filter- optional

Choose a subset of your data source to focus your data stream and reduce noisy data.



{ "terms": { "rule.groups": ["web"] } } ×

+ Add data filter

Timestamp

Select the time field you want to use for the time filter.

Timestamp field

Choose the time field you want to use for time filter.

@timestamp ×

W.

Anomaly detect...

Detectors

Create detector

Operation settings

Detector interval

Define how often the detector collects data to generate anomalies. The shorter the interval is, the more real time the detector results will be, and the more computing resources the detector will need. [Learn more](#)

1

minutes

Window delay

Specify a window of delay for a detector to fetch data, if you need to account for extra processing time. [Learn more](#)

1

minutes

Custom result index

Store detector results to your own index. [Learn more](#)

☒ Enable custom result index

⚠ You can't change the custom result index after creating the detector.

Field

opensearch-ad-plugin-result- http-anomaly

Custom result index name must contain less than 255 characters including the prefix "opensearch-ad-plugin-result-". Valid characters are a-z, 0-9, -(hyphen) and _(underscore).

☒ Enable flattened custom result index

Flattening the custom result index will make it easier to query them on the dashboard. It also allows you to perform term aggregations on categorical fields.

☒ Enable custom result index lifecycle management

Min Index Age - optional

This setting would define a specific threshold for the age of an index. When this threshold is surpassed, a rollover will be triggered automatically.

7

days

Min Index Size - optional

This setting would define a specific threshold for the size of an index. When this threshold is surpassed, a rollover will be triggered automatically.

51200

MB

Index TTL - optional

This setting would define the duration after which an index is considered expired and eligible for deletion.

60

days

W.

Anomaly detect...

Detectors

Create detector

3 Set up detector jobs

4 Review and create

Features

A feature is the field in your index that you use to check for anomalies. You can add up to 5 features. [Learn more](#)

⌵ http_error_count

Feature name

http_error_count

Enter a descriptive, unique name. The name must contain 1-64 characters. Valid characters are a-z, A-Z, 0-9, -(hyphen) and _(underscore).

Feature state

☒ Enable feature

Find anomalies based on

Field value

Aggregation method

The aggregation method determines what constitutes an anomaly.

count()

E.g, if you choose min(), the detector focuses on finding anomalies based on the minimum values of your feature.

Field

rule.id

Anomaly criteria

Acceptable difference between the expected and actual values

Deviation in any direction (default)

> Customize Suppression Rules

Advanced settings Hide

Shingle size

Set the number of intervals to consider in a detection window for your model. The anomaly detector expects the shingle size to be in the range of 1 and 128. The default shingle size is 8. We recommend that you don't choose 1 unless you have two or more features. Smaller values might increase recall but also false positives. Larger values might be useful for ignoring noise in a signal. [Learn more](#)

intervals

Sparse data handling

Choose how to handle missing data points. [Learn more](#)

STEP 2: DETECTOR REVIEW & JOB SETUP

What is being done:

- Reviewed detector configuration
- Enabled **real-time detection**
- Enabled **historical analysis** for baseline learning

Detectors

Create detector

Sample anomalies

You can preview anomalies based on sample feature input

Use the sample data as a reference to fine tune settings. To see the latest preview with your adjustments, click "Refresh preview". Once you are done with your edits, save your changes and run the detector to see real time anomalies for the new data set. [Learn more](#)

Refresh preview

Sample anomaly history

last 7 days

Show dates

Sample anomaly occurrences

0

Sample anomaly grade

0

Sample confidence

0

Last sample anomaly occurrence

-

Sample confidence

Sample anomaly grade

Sample feature breakdown

http_error_count

Field rule id: Aggregation method: value_count State: Enabled

Edit feature

Sample feature output

Set up detector jobs

Real-time detection

Real-time detection lets you find anomalies in your data in near real-time. To receive accurate and real-time anomalies, the detector needs to start and collect sufficient data to include your latest changes. The earlier the detector starts running, the sooner the real-time anomalies will be available. [Learn more](#)

☒ Start real-time detector automatically (recommended)

Historical analysis detection

Historical analysis detection lets you analyze and apply machine learning models over long historical data windows (weeks or months). You can identify anomaly patterns, seasonality, and trends. [Learn more](#)

☒ Run historical analysis detection

Historical analysis date range

last 30 days

Show dates

Select a date range for your historical analysis (you may adjust later).

Detector settings

Edit

We identified some areas that might improve your model

Source index data is potentially too sparse for model training. Consider changing interval length or ingesting more data

Name	Data source index	Data filter	Detector interval	Description	Timestamp	Window delay	Custom result index
HTTP_Error_Anomaly	wazuh-alerts-*	Custom expression: View code	1 Minutes	Detects abnormal spikes in HTTP error responses (4xx/5xx) from Wazuh alerts to identify potential web attacks, application failures, or DoS attempts.	@timestamp	1 Minutes	opensearch-ad-plugin-result-http-anomaly
Flatten custom result index	Custom result index min age	Custom result index min size	Custom result index TTL				
Yes	7 Days	51200 MB	60 Days				

Model configuration

Edit

Model configurations are validated

Additional settings

Category field	Shingle size	Imputation method	Custom values
-	8	ignore	-

Features (1)

Feature name	Feature definition	Feature state	Anomaly Criteria
http_error_count	Field: rule.id Aggregation method: value_count	Enabled	any

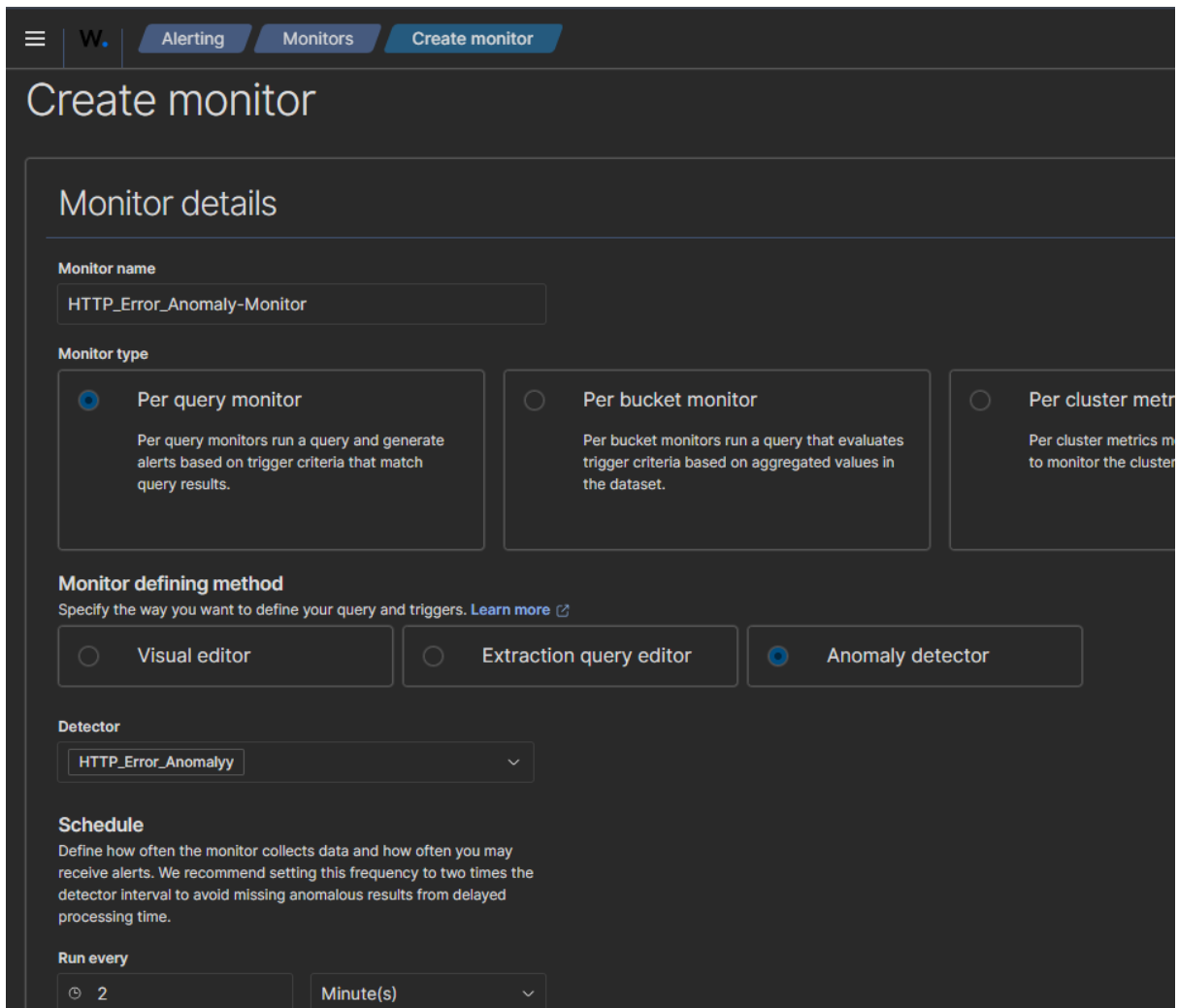
STEP 3: ALERTING & SLACK INTEGRATION

Monitor & Trigger Configuration:

- **Monitor type:** Anomaly Detector
- **Trigger name:** HTTP_Anomaly_HighErrorRate
- **Severity:** High
- **Trigger condition:**
 - Anomaly Grade > 0.7
 - Confidence > 0.7

Action:

- Slack notification sent to SOC channel



Create monitor

Monitor details

Monitor name

HTTP_Error_Anomaly-Monitor

Monitor type

- ☒ **Per query monitor**
Per query monitors run a query and generate alerts based on trigger criteria that match query results.
- ☐ **Per bucket monitor**
Per bucket monitors run a query that evaluates trigger criteria based on aggregated values in the dataset.
- ☐ **Per cluster metric monitor**
Per cluster metrics monitor the cluster.

Monitor defining method
Specify the way you want to define your query and triggers. [Learn more](#)

- ☐ Visual editor
- ☐ Extraction query editor
- ☒ Anomaly detector

Detector

HTTP_Error_Anomaly

Schedule
Define how often the monitor collects data and how often you may receive alerts. We recommend setting this frequency to two times the detector interval to avoid missing anomalous results from delayed processing time.

Run every

2 Minute(s)

Triggers (1)

⌵ HTTP_Anomaly_HighErrorRate

Trigger name

HTTP_Anomaly_HighErrorRate

Severity level

2 (High)

Trigger type

Define type of anomaly detector trigger

Anomaly detector grade and confidence

Anomaly grade threshold

IS ABOVE

0.7

Anomaly confidence threshold

IS ABOVE

0.7

Actions (1)

Define actions when trigger conditions are met.

⌵ Notification: Notify_SOC_HTTP_Anomaly

Action name

Notify_SOC_HTTP_Anomaly

Names can only contain letters, numbers, and special characters

Channels

[Channel] soc-slack-alerts

Manage channels [↗](#)

Message subject

Alerting Notification action

Message

Embed variables in your message using Mustache templates. [Learn more](#) [↗](#)

Monitor {{ctx.monitor.name}} just entered alert status. Please investigate the issue.

- Trigger: {{ctx.trigger.name}}

- Severity: {{ctx.trigger.severity}}

- Period start: {{ctx.trigger.start}} UTC

☒ Preview message

Message preview

Monitor HTTP_Error_Anomaly-Monitor just entered alert status. Please investigate the issue.

- Trigger: HTTP_Anomaly_HighErrorRate

- Severity: 2

- Period start: 2026-01-16T05:42:36Z UTC

- Period end: 2026-01-16T05:44:36Z UTC

✂ STEP 4: ATTACK SIMULATION FROM KALI

What is being done:

- Simulated HTTP error flood using Kali Linux
- Generated大量 404 Not Found requests using curl loop

This caused:

- Abnormal spike in HTTP error rate
- Deviation from learned baseline

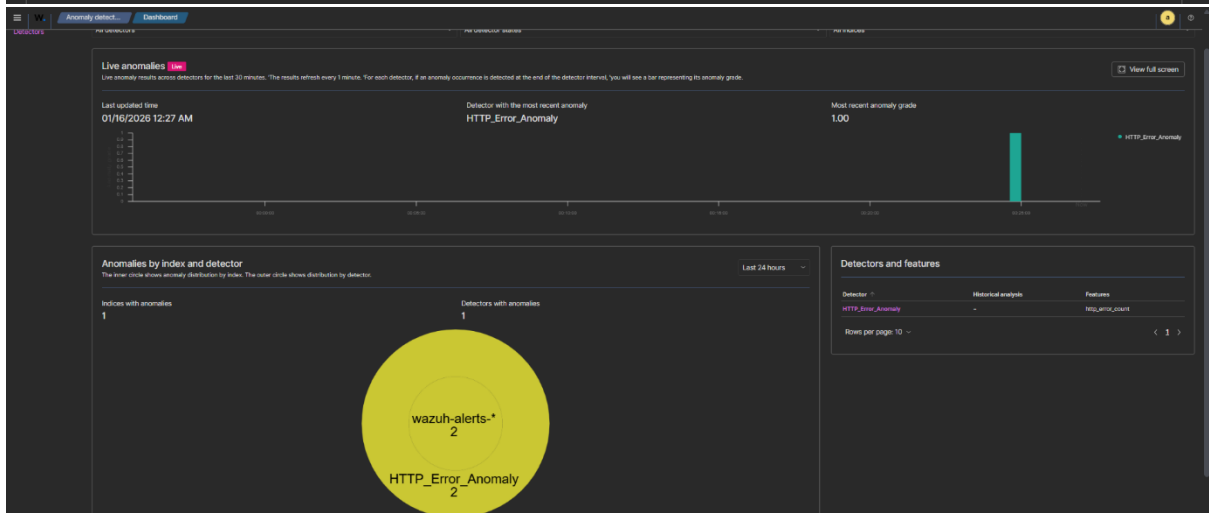
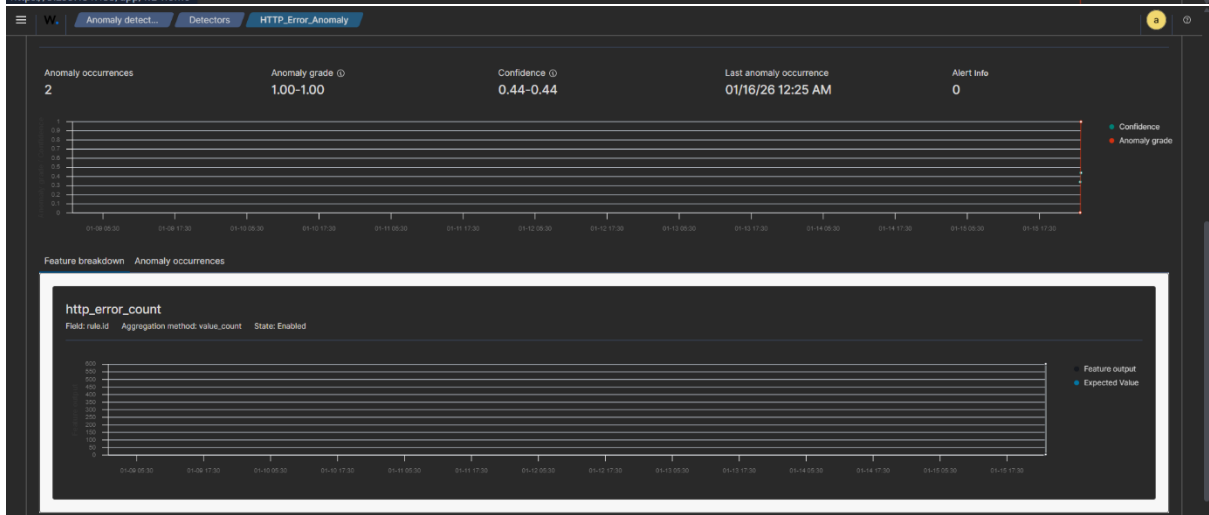
```
(kali@kali)-[~]
$ for i in {1..300}; do
  curl http://          /test$i > /dev/null
done
```

% Total	% Received	% Xferd	Average Dload	Average Upload	Speed	Time Total	Time Spent	Time Left	Current Speed
100	272	100	272	0	0	79416	0	--:--:--	136000
% Total	% Received	% Xferd	Average Dload	Average Upload	Speed	Time Total	Time Spent	Time Left	Current Speed
100	272	100	272	0	0	86596	0	--:--:--	90666
% Total	% Received	% Xferd	Average Dload	Average Upload	Speed	Time Total	Time Spent	Time Left	Current Speed
100	272	100	272	0	0	113238	0	--:--:--	136000
% Total	% Received	% Xferd	Average Dload	Average Upload	Speed	Time Total	Time Spent	Time Left	Current Speed
100	272	100	272	0	0	167590	0	--:--:--	272000
% Total	% Received	% Xferd	Average Dload	Average Upload	Speed	Time Total	Time Spent	Time Left	Current Speed
100	272	100	272	0	0	109943	0	--:--:--	136000
% Total	% Received	% Xferd	Average Dload	Average Upload	Speed	Time Total	Time Spent	Time Left	Current Speed
100	272	100	272	0	0	222585	0	--:--:--	272000
% Total	% Received	% Xferd	Average Dload	Average Upload	Speed	Time Total	Time Spent	Time Left	Current Speed
100	272	100	272	0	0	175257	0	--:--:--	272000
% Total	% Received	% Xferd	Average Dload	Average Upload	Speed	Time Total	Time Spent	Time Left	Current Speed

STEP 5: ANOMALY DETECTED (ML OUTPUT)

What happened:

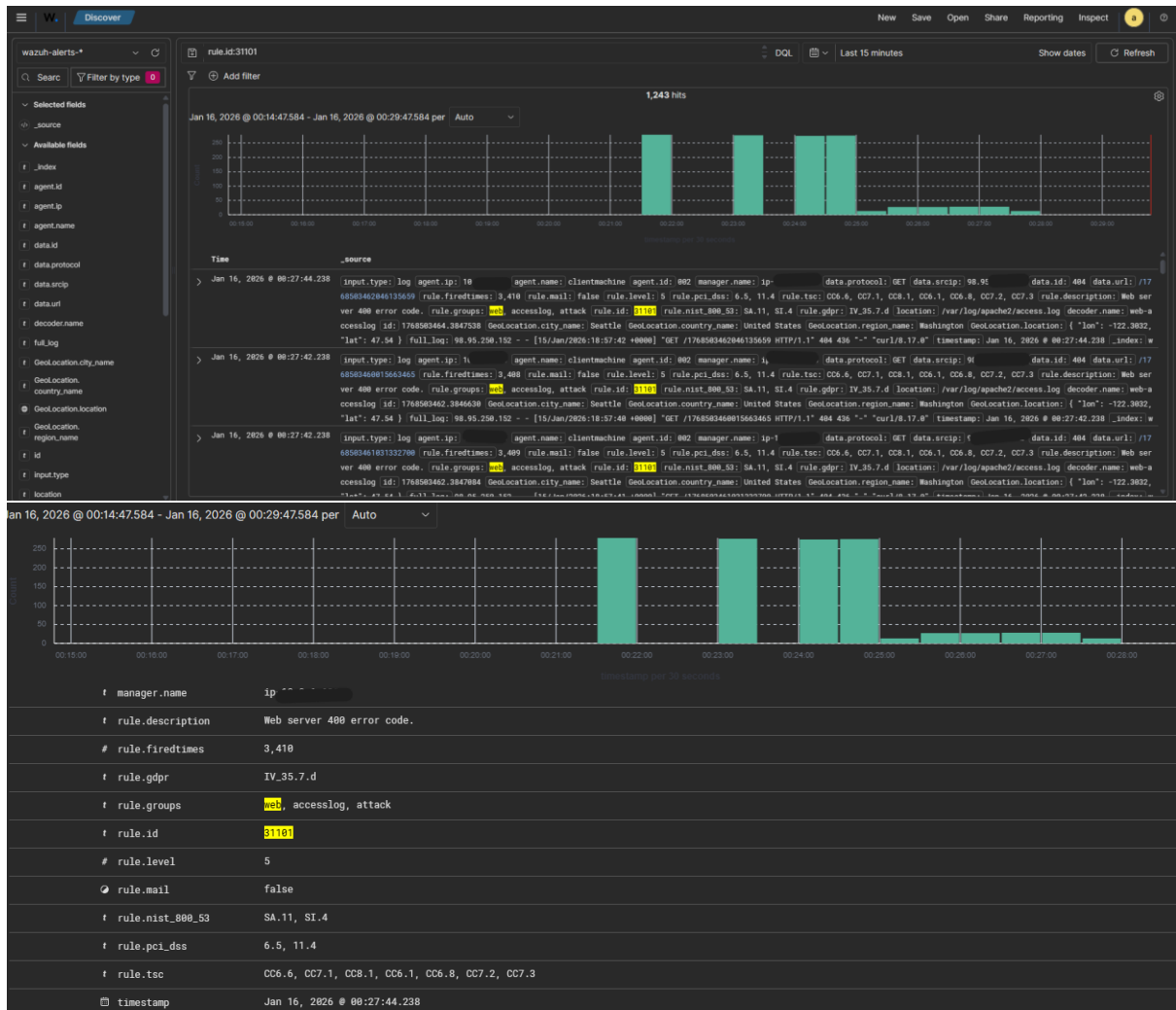
- OpenSearch ML detected abnormal behavior
- Anomaly grade reached **1.00**
- Detector displayed **red anomaly bar**
- Live anomaly visible in dashboard



STEP 6: LOG INVESTIGATION (DISCOVER)

Investigation actions:

- Filtered logs by:
 - rule.groups: web
 - rule.id
- Identified repeated requests from attacker IP
- Observed excessive 404 responses



Mapping:

- **Tactic:** Reconnaissance

- **Technique:**

- **T1595 – Active Scanning**
- **T1046 – Network Service Discovery**

The attacker behavior matched **web enumeration activity**.



103 hits						
Jan 16, 2026 @ 00:17:23.558 - Jan 16, 2026 @ 00:32:23.558						
Export Formatted Reset View 964 available fields Columns Density 1 fields sorted Full screen						
timestamp	agent.name	rule.mitre.id	rule.mitre.tactic	rule.description	rule.level	rule.id
Jan 16, 2026 @ 00:27:38.2...	clientmachine	T1595.002	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:27:22.2...	clientmachine	T1595.002	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:27:08.2...	clientmachine	T1595.002	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:26:54.2...	clientmachine	T1595.002	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:26:40.2...	clientmachine	T1595.002	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:26:26.2...	clientmachine	T1595.002	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:26:12.2...	clientmachine	T1595.002	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:25:58.2...	clientmachine	T1595.002	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:25:44.2...	clientmachine	T1595.002	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:25:30.2...	clientmachine	Navigate to MITRE ATT&C - Intelligence and see the technique details	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:24:52.5...	clientmachine		Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:24:52.5...	clientmachine	T1595.002	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151
Jan 16, 2026 @ 00:24:52.4...	clientmachine	T1595.002	Reconnaissance	Multiple web server 400 error codes from same source ip.	10	31151

[illegible]

MITRE ATTACK

Dashboard	Intelligence	Framework	Events	
Search in all resources Techniques (750)				
Groups	Search			
Mitigations	ID			
Software				
Tactics	T1002.008			
Techniques				
	T1548			
	T1134			

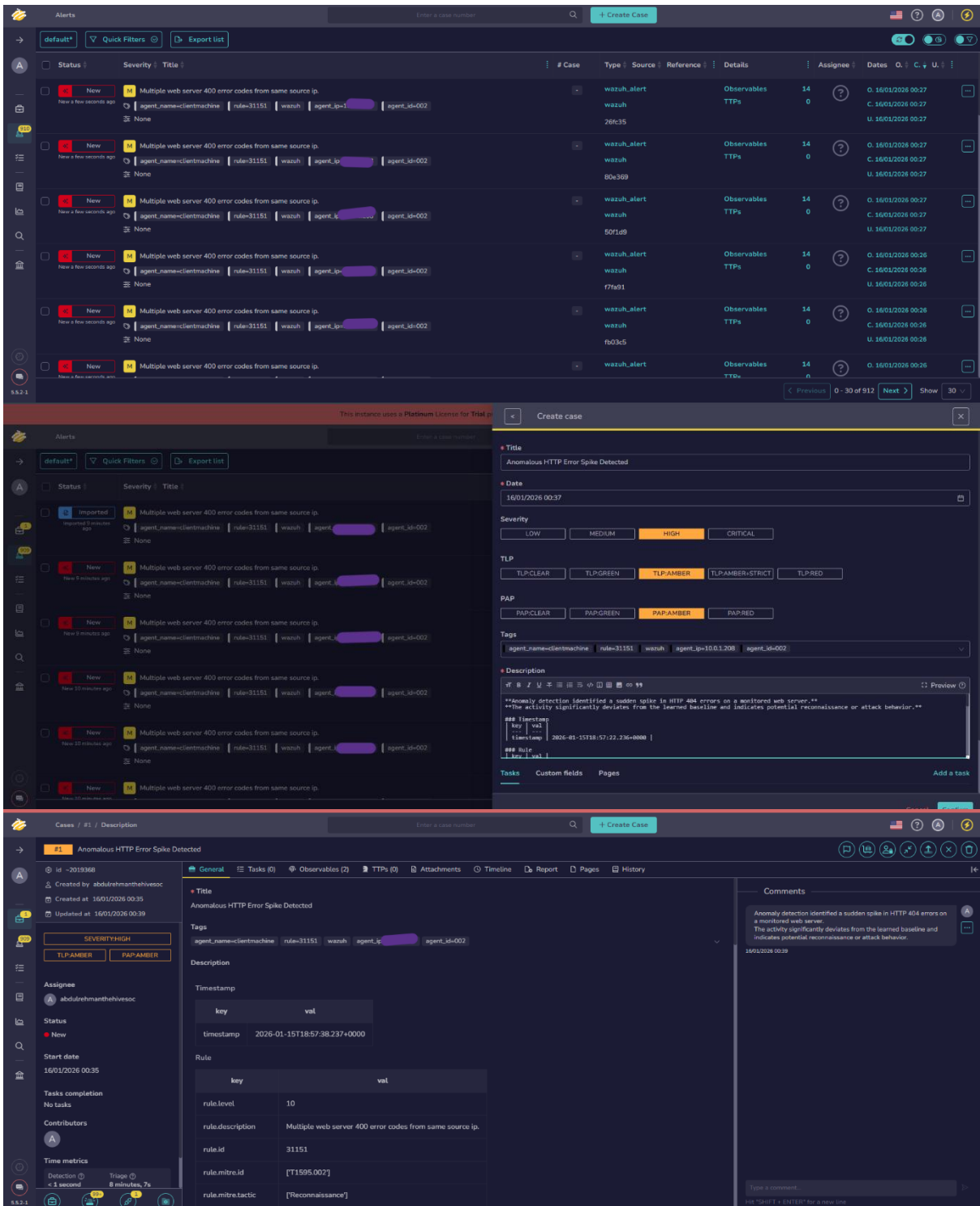
Details

ID	Name	Created Time	Modified Time	Version
T1595.002	Vulnerability Scanning	Oct 2, 2020 @ 22:25:16.047	Mar 14, 2023 @ 02:16:31.907	1.0
Description Adversaries may scan victims for vulnerabilities that can be used during targeting. Vulnerability scans typically check if the configuration of a target host/application (ex: software and version) potentially aligns with the target of a specific exploit the adversary may seek to use. These scans may also include more broad attempts to Gather Victim Host Information that can be used to identify more commonly known, exploitable vulnerabilities. Vulnerability scans typically harvest running software and version numbers via server banners, listening ports, or other network artifacts.(Citation: OWASP Vuln Scanning) Information from these scans may reveal opportunities for other forms of reconnaissance (ex: Search Open Websites/Domains or Search Open Technical Databases), establishing operational resources (ex: Develop Capabilities or Obtain Capabilities), and/or initial access (ex: Exploit Public-Facing Application).				
Groups				
ID	Name	Description		
G0123	Volatile Cedar	Volatile Cedar is a Lebanese threat group that has targeted individuals, companies, and institutions worldwide. Volatile Cedar has been operating since 2012 and is motivated by political and ideological interests.(Citation: CheckPoint Volatile Cedar March 2015)(Citation: ClearSky Lebanese Cedar Jan 2021)		
G0139	TeamTNT	TeamTNT is a threat group that has primarily targeted cloud and containerized environments. The group has been active since at least October 2019 and has mainly focused its efforts on leveraging cloud and container resources to deploy cryptocurrency miners in victim environments.(Citation: Palo Alto Black-T October 2020)(Citation: Lacework TeamTNT May 2021)(Citation: Interze TeamTNT September 2020)(Citation: Cado Security TeamTNT Worm August 2020)(Citation: Unit 42 Hologram Malware)(Citation: Trend Micro TeamTNT)(Citation: ATT TeamTNT Chinares September 2020)(Citation: Aqua TeamTNT August 2020)(Citation: Interze TeamTNT Explosion September 2021)		
	Sandworm Team	Sandworm Team is a destructive threat group that has been attributed to Russia's General Staff Main Intelligence Directorate (GRU) Main Center for Special Technologies (G15T1) military unit 74455.(Citation: US District Court Indictment GRU LHM 74455 October 2020)(Citation: UN HCSO Olympic Attacks October 2020) This group has been active since at least 2008.(Citation: Proton) Sandworm 2014(Citation: TrendMicro VooP000D.FHAR)(Citation: ISOD)		

STEP 8: INCIDENT RESPONSE IN THEHIVE

Actions performed:

- Created a **TheHive** case
- Added:
 - Title
 - Severity
 - Observables (IP, URL)
 - TTP mapping (MITRE)
- Assigned tasks:
 - Analysis
 - Containment
 - Mitigation



The screenshot displays the TheHive interface, showing a list of alerts and a detailed view of a specific case.

Alerts List:

Status	Severity	Title	# Case	Type	Source	Reference	Details	Assignee	Dates
New	High	Multiple web server 400 error codes from same source ip.		wazuh_alert			Observables	14	O: 16/01/2026 00:27
				wazuh			TTPs	0	C: 16/01/2026 00:27
					26fc35				U: 16/01/2026 00:27
New	High	Multiple web server 400 error codes from same source ip.		wazuh_alert			Observables	14	O: 16/01/2026 00:27
				wazuh			TTPs	0	C: 16/01/2026 00:27
					80e369				U: 16/01/2026 00:27
New	High	Multiple web server 400 error codes from same source ip.		wazuh_alert			Observables	14	O: 16/01/2026 00:27
				wazuh			TTPs	0	C: 16/01/2026 00:27
					501d9				U: 16/01/2026 00:27
New	High	Multiple web server 400 error codes from same source ip.		wazuh_alert			Observables	14	O: 16/01/2026 00:26
				wazuh			TTPs	0	C: 16/01/2026 00:26
					r7fa1				U: 16/01/2026 00:26
New	High	Multiple web server 400 error codes from same source ip.		wazuh_alert			Observables	14	O: 16/01/2026 00:26
				wazuh			TTPs	0	C: 16/01/2026 00:26
					fb03c5				U: 16/01/2026 00:26
New	High	Multiple web server 400 error codes from same source ip.		wazuh_alert			Observables	14	O: 16/01/2026 00:26
							TTPs	0	C: 16/01/2026 00:26

Case Details (Anomalous HTTP Error Spike Detected):

- Title:** Anomalous HTTP Error Spike Detected
- Date:** 16/01/2026 00:37
- Severity:** LOW, MEDIUM, **HIGH**, CRITICAL
- TLP:** TLP-CLEAR, TLP-GREEN, **TLP-AMBER**, TLP-AMBER-STRICT, TLP-RED
- PAP:** PAP-CLEAR, PAP-GREEN, **PAP-AMBER**, PAP-RED
- Tags:** agent_name=clientmachine, rule=31151, wazuh, agent_ip=10.0.1.208, agent_id=002
- Description:**

```
*** Anomaly detection identified a sudden spike in HTTP 404 errors on a monitored web server.***  
***The activity significantly deviates from the learned baseline and indicates potential reconnaissance or attack behavior.***  
*** timestamp  
key | val |  
timestamp | 2026-01-15T18:57:38.237+0000 |  
*** Rule  
key | val |
```
- Tasks:** Add a task

Case Summary (Anomalous HTTP Error Spike Detected):

- ID:** -2013368
- Created by:** abdulrehmanthehivevoc
- Created at:** 16/01/2026 00:35
- Updated at:** 16/01/2026 00:39
- Severity:** **SEVERITY-HIGH**
- TLP:** **TLP-AMBER**, PAP-AMBER
- Assignee:** abdulrehmanthehivevoc
- Status:** New
- Start date:** 16/01/2026 00:35
- Tasks completion:** No tasks
- Contributors:**
- Time metrics:** Detection: 4 seconds, Triage: 8 minutes, 7s

General Information:

- Title:** Anomalous HTTP Error Spike Detected
- Tags:** agent_name=clientmachine, rule=31151, wazuh, agent_ip=10.0.1.208, agent_id=002
- Description:**
- Timestamp:**
- key:** val
- timestamp:** 2026-01-15T18:57:38.237+0000
- Rule:**
- key:** val
- rule.level:** 10
- rule.description:** Multiple web server 400 error codes from same source ip.
- rule.id:** 31151
- rule.mitre.id:** [T1595.002]
- rule.mitre.tactic:** [Reconnaissance]

Comments:

- Anomaly detection identified a sudden spike in HTTP 404 errors on a monitored web server. The activity significantly deviates from the learned baseline and indicates potential reconnaissance or attack behavior.
- 16/01/2026 00:39

The top screenshot displays the 'Observables' tab in Splunk SOAR. The case is 'Anomalous HTTP Error Spike Detected' (ID: 2019368). It lists several observables with details like hostname, url, ip, and associated dates. The bottom screenshot displays the 'TTPs' tab for the same case, showing a single TTP entry: 'reconnaissance' with the technique 'T1595 - Active Scanning'.

STEP 9: MITIGATIONS & CONTAINMENT

Mitigation actions:

- Blocked attacker IP using:

```
sudo iptables -A INPUT -s <attacker_ip> -j DROP
```
- Restarted Apache services
- Enabled rate limiting & ModSecurity

```
ubuntu@ip-10.0.0.152:~$ sudo tail -n 5 /var/log/apache2/access.log
98.100.152.152 - - [15/Jan/2026:18:57:40 +0000] "GET /1768503460015663465 HTTP/1.1" 404 436 "-" "curl/8.17.0"
98.100.152.152 - - [15/Jan/2026:18:57:41 +0000] "GET /1768503461031332700 HTTP/1.1" 404 436 "-" "curl/8.17.0"
98.100.152.152 - - [15/Jan/2026:18:57:42 +0000] "GET /1768503462046135659 HTTP/1.1" 404 436 "-" "curl/8.17.0"
98.100.152.152 - - [15/Jan/2026:19:13:28 +0000] "GET /test999 HTTP/1.1" 404 436 "-" "curl/8.17.0"
98.100.152.152 - - [15/Jan/2026:19:14:47 +0000] "GET /test999 HTTP/1.1" 404 436 "-" "curl/8.17.0"
ubuntu@ip-10.0.0.152:~$ sudo iptables -A INPUT -s 98.100.152.152 -j DROP
ubuntu@ip-10.0.0.152:~$ sudo iptables -L -n --line-numbers
Chain INPUT (policy ACCEPT)
num target prot opt source destination
1 DROP 0 -- 10.0.0.0/0 0.0.0.0/0
2 DROP 0 -- 10.0.0.0/0 0.0.0.0/0
3 DROP 0 -- 98.100.152.152 0.0.0.0/0

Chain FORWARD (policy ACCEPT)
num target prot opt source destination

Chain OUTPUT (policy ACCEPT)
num target prot opt source destination
ubuntu@ip-10.0.0.152:~$ sudo a2enmod ratelimit
Considering dependency env for ratelimit:
Module env already enabled
Module ratelimit already enabled
ubuntu@ip-10.0.0.152:~$ sudo systemctl restart apache2
ubuntu@ip-10.0.0.152:~$
```

(clientmachine)

✅ STEP 10: VERIFICATION & CASE CLOSURE

Verification:

- Retried attack from Kali
- Server was unreachable / blocked
- No further anomalies observed

Case status:

- Incident contained
- Case closed in TheHive with resolution notes

The screenshot is divided into two main sections. The top section is a terminal window showing a command and its output:

```
(kali㉿ kali)-[~]  
$ curl http://52.147.100.100/test999  
  
curl: (28) Failed to connect to 52.147.100.100 port 80 after 135018 ms: Could not connect to server  
  
(kali㉿ kali)-[~]  
$
```

The bottom section shows the TheHive web interface. The main view displays a list of cases, with case #1 selected. The case details for #1 are shown on the right, including its status (True Positive), severity (High), and a summary of the incident. The summary text reads: "Response actions applied: - Source IP blocked at host firewall - Web application firewall enabled - Monitoring confirmed no further malicious activity". The interface also shows the case's timeline, tasks, and observables.

🚀 PROJECT OUTCOME

- Successfully detected abnormal web traffic using ML
- Alerted SOC team in real time
- Investigated logs and attacker behavior
- Mapped activity to MITRE ATT&CK
- Responded and mitigated threat
- Fully documented incident lifecycle

SKILLS GAINED

- ML-based anomaly detection (OpenSearch AD)
- SOC alerting & monitoring
- Log analysis & threat hunting
- MITRE ATT&CK mapping
- Incident response using TheHive
- Linux host-based mitigation
- End-to-end SOC workflow execution

FINAL CONCLUSION

This project demonstrates **practical SOC analyst capabilities**, combining **behavior-based detection, alerting, investigation, and incident response**.

It reflects a **real enterprise SOC scenario**, not a lab-only setup, making it suitable for **portfolio, interviews, and resume projects**.
